Coursework Instruction

Topic: Development of a new sensing device or system

The aim of this coursework is to broaden your knowledge in sensors and instrumentation and to enhance your skill in academic writing.

A choice is to develop a portable capacitance metre, based on a charge/discharge capacitance measuring circuit. A relevant paper is available on Blackboard (under Course Content/Supporting material/The Coursework). Two guest lectures, which can be watched on Blackboard (under Course Content/Synchronous sessions), also help your coursework writing. You also have the option to choose your own topic about a sensing device or system, or based on your 3rd year individual project.

You need to consider

- Cost
- Functionality
- Performance and
- Manufacturability of the proposed device or system.

In your coursework, you need to include the following contents:

(1) **Abstract (10 marks):** One paragraph (150 words) summarising: aims and objectives; context and identified customer need; market factors and competition; idea, design and implementation.

(2) **Introduction and motivation (10 marks):** Brief statement of your idea about the sensing needs and motivation; identification of the customers; general appreciation of market and competitors; important specifications; and target market price.

(3) **Review of existing technology (20 marks):** Evidence of keyword literature search, e.g. through Google Scholar and database of patents (http://www.freepatentsonline.com); description of the state-of-the-art supported by references; summary of related sensor technologies; and assessment of the uniqueness of the proposed sensing device or system.

(4) **Description of the proposed product, including working principle, product design and development (40 marks):** Sensors and Instrumentation Page 2 of 2 Quantitative specification wherever possible (e.g. accuracy, sensitivity, precision, size, weight, robustness, etc.; design details (including e.g. part numbers, costs, interface specification, etc.); outline of system electronics and software (as appropriate) and any miscellaneous requirements (such as mechanical components); block diagram of sensing device or system architecture and/or software flowchart; packaging requirements and/or user interface details; estimation of performance. Construction requirements (equipment needed, special facilities, etc.); testing programme (performance, safety, etc.); estimated costs and time schedule (including materials, construction and testing).

(5) **Leaflet for marketing, showing benefits to manufacturer and customers (10 marks)**

(6) **References, in particular the websites, papers, etc. you have read, which are relevant to the proposed solution (10 marks)**

Other requirements:

- Full report should be limited to 6 pages, including your reference list and the leaflet but excluding appendices if you have.

- You should use the IEEE template for the report (except for the leaflet): The template is provided on Blackboard (under Course Content/Supporting material/The Coursework).

- Use preferred format for the 1-page leaflet.

- Submit your coursework as a single pdf file via Blackboard.

- Turnitin will check for plagiarism.